# 1 Introduction to scattering theory

Scattering experiments are among the most important tools in physics. From Rutherford's discovery of the atomic nucleus to the detection of the Higgs boson at the LHC, we learn about the fundamental structure of matter by shooting particles at targets and observing what comes out. In this section we set up the quantum-mechanical scattering problem, introduce the key concepts of spherical waves and scattering amplitudes, and connect the formalism to the time-dependent perturbation theory we developed earlier.

This section is inspired by Sakurai & Napolitano Chapter 6, Shankar Chapter 19, and at first, Bart's MIT Open Courseware notes. On the formalism part, we will recover several results in Sakurai, but we will not follow his presentation closely. We have our own technology to put to use.

More reading: once we see Lippman-Schwinger, you can read more about this topics in Sakurai's Chapters 6.4 to 6.9.

Recall the one-dimensional scattering problem you (probably) solved last semester. A particle incident on a potential step at $x = 0$ had its wavefunction decomposed into three pieces: an incoming plane wave $e^{ikx}$, a reflected wave $e^{-ikx}$, and a transmitted wave. The reflection and transmission coefficients $R$ and $T$ told us the probability of each outcome. They were defined by the asymptotic behavior of the wavefunction at $x \to \pm\infty$:

$$\psi(x) \xrightarrow{x\to-\infty} e^{ikx} + R\,e^{-ikx}, \qquad \psi(x) \xrightarrow{x\to+\infty} T e^{ikx}. \tag{1}$$

The solutions to the Schrödinger equation alongside matching conditions in the scattering region $x \approx 0$ determined $R$ and $T$ (see Shankar Sec. 5.4 for the solution of this problem. Compare, e.g., the form of Shankar's Eq. (5.4.11) with the form above.) This is the essence of scattering theory: we set up a problem with an incoming wave, solve the Schrödinger equation, and read off the coefficients of the outgoing waves at large distances to determine the scattering probabilities.

In one dimension, the physics of scattering is captured by two numbers: $R$ and $T$. The incoming state is a plane wave moving to the right, and the outgoing state is either a reflected wave moving to the left or a transmitted wave continuing to the right. In three dimensions, the situation is analogous. Let us describe the incoming particle by a plane wave $e^{i\vec{k}\cdot\mathbf{x}}$ with $\vec{k}$ along the $z$-axis. This wave encounters a localized potential $V(\mathbf{x})$ and scatters. The potential can scatter the particle in any direction, not just forward or backward. If the potential is spherically symmetric, the scattering will be azimuthally symmetric, but in general it can have a complicated angular dependence. Now, instead of a single transmission coefficient, we need a function that tells us the probability of scattering into each direction $(\theta, \phi)$. This will be the differential cross section, the same quantity we encountered in the time-dependent perturbation theory problems. Finding this function is the central motivation behind scattering theory.

But what do outgoing waves look like in three dimensions? What is reflected or transmitted in 3D? This is obviously much more complicated in the general case, but we can make better progress narrowing the scope down to potential scattering. Let us define the scope of the problem we want to solve:

- Particles are non-relativistic.

- The potential is localized in space: $V(\mathbf{x})$ is negligible outside some radius $a$.[1]
- The potential is static: it does not depend on time.
- The scattering is elastic: the particle does not change its internal state or produce new particles.
- We are only interested in the asymptotic behavior of the wavefunction at large distances $r \gg a$, where the potential is negligible and the scattered wave can be identified as a free solution of the Schrödinger equation.

There are also assumptions that we will make in a first moment, but could relax later on. These include

- The incoming state is a plane wave with definite momentum $\vec{k}$.
- The potential is spherically symmetric and central: $V(\mathbf{x}) = V(r)$
- The particles involved are spinless.

The central potential is a particularly simple case because the scattering does not mix different angular momenta $\ell$ modes. It is also quite common: the most elementary and useful application of the scattering on a hard sphere is a central potential. The same is true for the Coulomb and the Yukawa potentials which we will encounter later on.

> I had mentioned that scattering is just an application of time-dependent perturbation theory, so how do we reconcile this with the time-independence of $V(\mathbf{x})$? The time dependence comes in through the fact that the particle comes from far away, interacts with the potential, and then travels far away. Note that this picture *requires* that the particle be described by a wavefunction that is localized in space (a wavepacket, as opposed to a plane wave), so that it can be said to "come in" and "go out". In practice, however, the finite size of the wavepacket can be neglected in many applications (even if not all), and we can work with plane waves as an idealization.

Under these assumptions, the 3D scattering problem is well described by an incoming wave that scatters off to a superposition of outgoing waves in all directions. This superposition is called the **scattered wave**, and the function that encodes the amplitude of scattering into each direction is called the **scattering amplitude**.

### 1.1 Spherical waves

Consider the free-particle Schrödinger equation in three dimensions:

$$-\frac{\nabla^2\psi(\mathbf{x})}{2m} = E\psi(\mathbf{x}), \qquad E = \frac{k^2}{2m}, \tag{2}$$

where $k > 0$ is the wavenumber. We know that plane waves $e^{i\vec{k}\cdot\mathbf{x}}$ are solutions. Are there solutions with spherical symmetry?

Let us look for solutions of the form $\psi(\mathbf{x}) = \frac{u(r)}{r}$, where $r = |\mathbf{x}|$. To do so, we need the Laplacian acting on a function that depends only on $r$.

[1]Note this is barely true for the Coulomb potential, which falls off as $1/r$.

**The Laplacian in spherical coordinates.** The full Laplacian in spherical coordinates is

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\hat{L}_{\theta,\phi}, \tag{3}$$

where $\hat{L}_{\theta,\phi}$ is the angular part of the Laplacian (involving $\theta$ and $\phi$ derivatives), which you may recognize as proportional to the angular momentum operator, $\hat{L}_{\theta,\phi} = -\hat{L}^2$. For a function $f(r)$ that depends only on $r$, the angular part vanishes and we have

$$\nabla^2 f(r) = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right) = \frac{1}{r}\frac{d^2}{dr^2}(r\,f(r)). \tag{4}$$

This identity, which you should verify, is crucial: it tells us that $\nabla^2(u(r)/r) = \frac{1}{r}\frac{d^2u}{dr^2}$, reducing the 3D Laplacian to a 1D second derivative.

Substituting $\psi = u(r)/r$ into Eq. (2) and using Eq. (4):

$$-\frac{1}{2m}\frac{1}{r}\frac{d^2u}{dr^2} = \frac{k^2}{2m}\frac{u(r)}{r}, \tag{5}$$

which simplifies to

$$\frac{d^2u}{dr^2} = -k^2u(r). \tag{6}$$

This is the same equation as for a free particle in one dimension! The general solution is $u(r) = Ae^{ikr} + Be^{-ikr}$, giving

$$\psi(r) = A\,\frac{e^{ikr}}{r} + B\,\frac{e^{-ikr}}{r}. \tag{7}$$

These are spherical waves: $e^{ikr}/r$ is an outgoing wave (propagating radially outward) and $e^{-ikr}/r$ is an incoming wave (propagating radially inward). The $1/r$ prefactor ensures that the probability current through any sphere of radius $r$ is independent of $r$, just as probability conservation demands.

Compute the radial probability current $\mathbf{j_r} = \frac{\hat{\mathbf{r}}}{2mi}\left(\psi^*\frac{\partial\psi}{\partial r} - \psi\frac{\partial\psi^*}{\partial r}\right)$ for $\psi = \frac{e^{ikr}}{r}$, and show that the flux $\Phi = \int \mathbf{j_r}\,\mathbf{dA} = \int(\mathbf{j_r}\cdot\hat{\mathbf{r}})\,r^2\,d\Omega$ through a sphere of radius $r$ is independent of $r$. (Recall the unit sphere element of area is just $d\Omega = \sin\theta\,d\theta\,d\phi$.)

Note the close parallel with the 1D problem: the two signs of the exponent ($e^{+ikx}$ and $e^{-ikx}$) correspond to right-moving and left-moving waves, while $e^{ikr}$ and $e^{-ikr}$ correspond to outgoing and incoming waves, respectively. Note, however, that while in 1D the outgoing and incoming waves had constant amplitude, in 3D they fall off as $1/r$ since the wavefront spreads over a sphere of area $4\pi r^2$. To conserve probability, the amplitude must fall as $1/r$.

## 1.2 The scattering amplitude

We can now state the scattering problem precisely. Consider a localized potential $V(\mathbf{x})$ that is negligible outside some radius $a$ (the "range" of the potential). We send in a plane wave along the $z$-axis,

$$\psi_i = e^{ikz}. \tag{8}$$

At large distances $r \gg a$, we expect the wavefunction to be a superposition of the incident plane wave and an outgoing spherical wave, and so we use the ansatz

$$\boxed{\psi(\mathbf{x}) \xrightarrow{r\gg a} \psi_i(\mathbf{x}) + \psi_s(\mathbf{x}) = e^{ikz} + f(\theta,\phi)\,\frac{e^{ikr}}{r}.} \tag{9}$$

The function $f(\theta, \phi)$ is called the **scattering amplitude**. It encodes all the information about how the potential deflects the incoming particle. Note that $k$ is the same in the incident and scattered waves, since we are assuming elastic scattering (i.e., the particle does not lose energy in the scattering process).

> Why must the scattered wave be an *outgoing* spherical wave, as opposed to an incoming one? Physically, we set up the problem with particles coming in from $z = -\infty$. After interacting with the potential, particles fly outward in all directions. There is no source of particles at infinity that would produce an incoming spherical wave converging on the target. This is just a boundary condition for our scattering problem.

The differential cross section $d\sigma/d\Omega$ is defined as the ratio of the scattered flux into solid angle $d\Omega$ to the incident flux per unit area. From Eq. (8), the incident flux density is just

$$\mathbf{j_i} = \frac{1}{2mi}\left(\psi_i^* \nabla \psi_i - \psi_i \nabla \psi_i^*\right) = \frac{k}{m}\hat{z}, \qquad j_i = |\mathbf{j_i}| = \frac{k}{m}. \tag{10}$$

The scattered radial flux through a surface element $r^2\, d\Omega$ at large $r$ is just

$$\mathbf{j_s} = \frac{1}{2mi}\left(\psi_s^* \nabla \psi_s - \psi_s \nabla \psi_s^*\right) \sim \frac{1}{2mi}\frac{|f|^2}{r^2}\left(\nabla(e^{ikr}) - \nabla(e^{-ikr})\right) = \frac{k}{m}\frac{|f(\theta,\phi)|^2}{r^2}\hat{r}. \tag{11}$$

where the tilde indicates we are dropping the incident wave contribution, which is negligible at large $r$ (though the forward direction $\theta = 0$ is a special case that we will discuss later on). With this, we can compute the number of particles scattered into solid angle $d\Omega$ per unit time (recall, the probability current density $\mathbf{j}$ is the number of particles crossing a unit area per unit time):

$$dN_s = j_s\, r^2\, d\Omega = \frac{k}{m}\frac{|f(\theta,\phi)|^2}{r^2}\, r^2\, d\Omega = \frac{k}{m}|f(\theta,\phi)|^2\, d\Omega. \tag{12}$$

Also, we know that the cross section is the the quantity we multiply by the incident flux to get the number of scattered particles, so using

$$dN_s = j_i\, d\sigma = \frac{k}{m}\, d\sigma, \tag{13}$$

therefore

$$\boxed{\frac{d\sigma}{d\Omega} = |f(\theta,\phi)|^2.} \tag{14}$$

This is a beautifully simple result: the differential cross section is just the modulus squared of the scattering amplitude. This $f(\theta, \phi)$ is the central object we need to compute. Note that we can also define a total cross section,

$$\sigma = \int d\Omega\, \frac{d\sigma}{d\Omega} = \int d\Omega\, |f(\theta,\phi)|^2. \tag{15}$$

> Our Eq. (14) is analogous to Sakurai's Eq. (6.59). Note that Sakurai uses the notation $f(\vec{k}', \vec{k})$ for the scattering amplitude as a function of the initial and final momenta, while we use $f(\theta, \phi)$ as a function of the scattering angles.

In what follows we will do a little of two things: 1) develop the formalism to more rigorously derive the scattering amplitude $f(\theta, \phi)$, and 2) rewrite and expand $f(\theta, \phi)$ in various ways to make it easier to compute for different potentials. Because some of the forms of $f(\theta, \phi)$ we will derive are quite general and powerful, we will derive them first. With that, we will also be able to connect the scattering amplitude to the time-dependent perturbation theory we developed in previous sections, and see how the Born approximation emerges as a special case.

### 1.3 The Partial Wave Expansion

Let us consider the case of a central potential, $V(\mathbf{x}) = V(r)$. Note that the scattering amplitude (and, therefore, the wavefunction) is not necessarily azimuthally symmetric, since the incident plane wave breaks spherical symmetry. The problem is symmetric under rotations around the $z$-axis (the wavefunction has no $\phi$ dependence), but not under $\theta$ rotations. As we will see, however, it is still useful and physically-insightful to work in spherical coordinates. As it turns out, the incoming plane wave can be written as a superposition of spherical waves, which are the natural solutions to the free Schrödinger equation in spherical coordinates. For each spherical wave (which will correspond to a specific angular momentum quantum number $\ell$), the scattering problem becomes much easier to solve, and the scattering amplitude $f(\theta, \phi)$ can be expressed in terms of the phase shifts of the spherical waves.

For now, we take $V = 0$ for a particle of energy $E = k^2/2m$. Such a free particle can be described by the wavefunction in spherical coordinates

$$\psi(\mathbf{x}) = \frac{u(r)}{r} Y_{\ell m}(\Omega). \tag{16}$$

Inserted it into Schrödinger's equation, gives

$$\frac{d^2u}{dr^2} + \left(k^2 - \frac{\ell(\ell+1)}{r^2}\right) u = 0. \tag{17}$$

In terms of the auxiliary radial variable[2]

$$\boxed{\rho = kr,} \tag{18}$$

this becomes

$$\frac{d^2u}{d\rho^2} + \left(1 - \frac{\ell(\ell+1)}{\rho^2}\right) u = 0. \tag{19}$$

This has solutions

$$u_\ell(\rho) = \rho j_\ell(\rho), \quad u_\ell(\rho) = \rho n_\ell(\rho), \tag{20}$$

where $j_\ell(\rho)$ and $n_\ell(\rho)$ are the spherical Bessel functions of the first and second kind, respectively. Recall these can be found recursively with

$$j_\ell(\rho) = (-\rho)^\ell \left(\frac{1}{\rho}\frac{d}{d\rho}\right)^\ell \left(\frac{\sin\rho}{\rho}\right), \qquad n_\ell(\rho) = -(-\rho)^\ell \left(\frac{1}{\rho}\frac{d}{d\rho}\right)^\ell \left(\frac{\cos\rho}{\rho}\right). \tag{21}$$

For reference, the first few spherical Bessel functions are given in Tables 1 and 2, alongside their expansions at small and large $\rho$. Most importantly, at large radii they both behave as spherical waves. Removing the $\rho$ prefactor, we have

$$j_\ell(kr) \sim \sin\left(kr - \frac{\ell\pi}{2}\right) = \frac{1}{kr}\left(\frac{e^{i(kr-\ell\pi/2)} - e^{-i(kr-\ell\pi/2)}}{2i}\right), \tag{22}$$

$$n_\ell(kr) \sim -\cos\left(kr - \frac{\ell\pi}{2}\right) = \frac{1}{kr}\left(\frac{e^{i(kr-\ell\pi/2)} + e^{-i(kr-\ell\pi/2)}}{2}\right). \tag{23}$$

Indeed, the spherical Bessel functions are just linear combinations of incoming and outgoing spherical waves, with a phase shift of $\ell\pi/2$. However, we note something important: at $r \to 0$, $j_\ell(kr) \sim (kr)^\ell$ is regular, while $n_\ell(kr) \sim -(2\ell-1)!!/(kr)^{\ell+1}$ is singular. Cleary our plane wave should not have a singularity at the origin, and the physics picks the regular solutions $j_\ell(kr)$.

[2]This is a very common combination of variables in scattering problems. Physically, $\rho = kr$ is the number of wavelengths contained in a radius $r$. As we will see, it will be instructive to study the limit of short-wavelength scattering ($\rho \gg 1$) and long-wavelength scattering ($\rho \ll 1$) when $r \sim a$ is the typical size of the scatterer.

| $\ell$ | $\rho\, j_\ell$ (exact) | $\rho\, j_\ell$ $(\rho \to 0)$ | $\rho\, j_\ell$ $(\rho \to \infty)$ |
|---|---|---|---|
| 0 | $\sin\rho$ | $\rho$ | $\sin\rho$ |
| 1 | $\dfrac{\sin\rho}{\rho} - \cos\rho$ | $\dfrac{\rho^2}{3}$ | $-\cos(\rho)$ |
| 2 | $\left(\dfrac{3}{\rho^2} - 1\right)\sin\rho - \dfrac{3}{\rho}\cos\rho$ | $\dfrac{\rho^3}{15}$ | $-\sin(\rho)$ |

Table 1: Spherical Bessel functions of the first kind, $j_\ell(\rho)$, for $\ell = 0, 1, 2$.

| $\ell$ | $\rho\, n_\ell$ (exact) | $\rho\, n_\ell$ $(\rho \to 0)$ | $\rho\, n_\ell$ $(\rho \to \infty)$ |
|---|---|---|---|
| 0 | $-\cos\rho$ | $-1$ | $-\cos\rho$ |
| 1 | $-\dfrac{\cos\rho}{\rho} - \sin\rho$ | $-\dfrac{1}{\rho}$ | $-\sin(\rho)$ |
| 2 | $-\left(\dfrac{3}{\rho^2} - 1\right)\cos\rho - \dfrac{3}{\rho}\sin\rho$ | $-\dfrac{3}{\rho^2}$ | $\cos(\rho)$ |

Table 2: Spherical Bessel functions of the second kind, $n_\ell(\rho)$, for $\ell = 0, 1, 2$.

From now on, we can drop the $n_\ell$ solutions altogether. In particular, it will be useful to know the integral representation of the spherical Bessel functions:

$$j_\ell(\rho) = \frac{1}{2}\int_{-1}^{1} e^{i\rho x} P_\ell(x)\, dx, \tag{24}$$

where $P_\ell(x)$ are the Legendre polynomials.

With these facts, let us return to the incoming plane wave $e^{ikz}$. We decompose it into spherical waves using the expansion

$$e^{ikz} = e^{ikr\cos\theta} = \sum_{\ell=0}^{\infty} a_\ell j_\ell(kr) P_\ell(\cos\theta), \tag{25}$$

where $P_\ell(\cos\theta)$ are the Legendre polynomials and $a_\ell$ are coefficients to be determined. This is called the **Rayleigh expansion** of a plane wave, or the **spherical wave expansion** of a plane wave.

Derivation: to determine these coefficients, we will explore the orthogonality of the Legendre polynomials:

$$\int_{-1}^{1} P_\ell(x)P_{\ell'}(x)\,dx = \frac{2}{2\ell+1}\,\delta_{\ell\ell'}. \tag{26}$$

Multiplying both sides of Eq. (25) by $P_{\ell'}(\cos\theta)\sin\theta$ and integrating over $\theta$ from 0 to $\pi$:

$$\begin{aligned}\int_0^\pi e^{ikr\cos\theta}P_{\ell'}(\cos\theta)\sin\theta\,d\theta &= \sum_{\ell=0}^{\infty} a_\ell j_\ell(kr)\int_0^\pi P_\ell(\cos\theta)P_{\ell'}(\cos\theta)\sin\theta\,d\theta \\ &= \sum_{\ell=0}^{\infty} a_\ell j_\ell(kr)\int_{-1}^{1} P_\ell(x)P_{\ell'}(x)\,dx \\ &= \sum_{\ell=0}^{\infty} a_\ell j_\ell(kr)\frac{2}{2\ell+1}\,\delta_{\ell\ell'} = a_{\ell'}j_{\ell'}(kr)\frac{2}{2\ell'+1}.\end{aligned} \tag{27}$$

On the other hand, the left-hand side can be evaluated using Eq. (24),

$$\int_0^\pi e^{ikr\cos\theta}P_{\ell'}(\cos\theta)\sin\theta\,d\theta = 2i^{\ell'}j_{\ell'}(kr). \tag{28}$$

Equating the two results gives $a_\ell = i^\ell(2\ell+1)$.

This is a special case of the more general expansion:

$$e^{i\vec{k}\cdot\mathbf{x}} = 4\pi\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} i^\ell j_\ell(kr)Y^*_{\ell m}(\hat{k})Y_{\ell m}(\hat{r}), \tag{29}$$

where $\hat{k} = \vec{k}/k$ and $\hat{r} = \mathbf{x}/r$ are the unit vectors in the directions of the incident wave and the observation point, respectively. Using Eq. (25) and comparing it to Eq. (16), we see that the incoming plane wave is a superposition of spherical waves with all angular momenta $\ell$, but only $m = 0$ (since the incident wave is azimuthally symmetric). Using the definition of the spherical harmonics, $Y_{\ell 0}(\theta,\phi) = \sqrt{\frac{2\ell+1}{4\pi}}P_\ell(\cos\theta)$, into Eq. (25) we can write the incoming plane wave as

$$e^{ikz} = \sum_{\ell=0}^{\infty} i^\ell\sqrt{4\pi(2\ell+1)}\,j_\ell(kr)Y_{\ell 0}(\theta,\phi). \tag{30}$$

This is a very important result: it tells us that the incoming plane wave can be expressed as a superposition of spherical waves with different angular momenta $\ell$. The partial wave expansion above is most useful for solving scattering problems with central potentials.

## 1.4 Phase Shifts

Now comes the important application of our expansion. When we turn on the potential $V(r)$, the spherical waves will be distorted by the scattering process. At large distances $r \gg a$, where the potential is negligible, the wavefunction must again take the form of a superposition of incoming and outgoing spherical waves. The difference is that the outgoing wave will now have a phase shift $\delta_\ell$ relative to the incoming wave, which encodes the effect of the potential on the scattering process.

To see this, we write the wavefunction at large $r$ using Eq. (22):

$$\psi(r) \xrightarrow{r\gg a} \sum_{\ell=0}^{\infty} i^\ell\sqrt{4\pi(2\ell+1)}\frac{1}{2i}\left(\frac{e^{i(kr-\ell\pi/2+2\delta_\ell)} - e^{-i(kr-\ell\pi/2)}}{2i}\right)P_\ell(\cos\theta). \tag{31}$$

Comparing this to the free case, we see that the effect of the potential is to introduce a phase shift $\delta_\ell$ in the outgoing spherical wave for each angular momentum $\ell$.

**Why only a phase shift?** At large $r$, the radial wavefunction for each partial wave is a superposition of incoming and outgoing spherical waves:

$$u_\ell(r) \equiv rR_\ell(r) \xrightarrow{r\to\infty} A_\ell^{\text{in}}\, e^{-ikr} + A_\ell^{\text{out}}\, e^{+ikr}. \tag{32}$$

The radial probability current for the $\ell$-th partial wave is

$$j_r^{(\ell)} = \frac{1}{2im}\left(u_\ell^* \frac{du_\ell}{dr} - u_\ell \frac{du_\ell^*}{dr}\right) = \frac{k}{m}\left(|A_\ell^{\text{out}}|^2 - |A_\ell^{\text{in}}|^2\right). \tag{33}$$

Since the potential is central, it does not mix different $\ell$ values, so probability conservation must hold independently for each partial wave. This requires $j_r^{(\ell)} = 0$, i.e., the outgoing flux must equal the incoming flux:

$$|A_\ell^{\text{out}}|^2 = |A_\ell^{\text{in}}|^2 \quad\Longrightarrow\quad A_\ell^{\text{out}} = e^{2i\delta_\ell}\, A_\ell^{\text{in}}. \tag{34}$$

In other words, the potential can only rotate the relative phase between the incoming and outgoing parts of each partial wave, and not their amplitude. The free Rayleigh expansion (Eq. (25)) fixes $A_\ell^{\text{in}}$, so the full asymptotic wavefunction becomes

$$u_\ell(r) \;\to\; \frac{1}{2i}\left(e^{2i\delta_\ell}\, e^{i(kr-\ell\pi/2)} - e^{-i(kr-\ell\pi/2)}\right) = e^{i\delta_\ell} \sin(kr - \ell\pi/2 + \delta_\ell), \tag{35}$$

confirming that $\delta_\ell$ is the phase shift of the radial standing wave pattern relative to the free solution.

Indeed, equating the outgoing part of the wavefunction to the scattering ansatz in Eq. (9) gives

$$\boxed{f(\theta) = \frac{1}{k}\sum_{\ell=0}^{\infty}(2\ell+1)e^{i\delta_\ell}\sin\delta_\ell P_\ell(\cos\theta).} \tag{36}$$

This is a powerful result: it tells us that if we can compute the phase shifts $\delta_\ell$ for each angular momentum $\ell$, we can determine the scattering amplitude and, therefore, the differential cross section. Note that the assumption of a central potential comes in here as the assumption that the scattering does not mix different $\ell$ values. This is why we can talk about "the $\ell = 0$ amplitude" or "the $\ell = 1$ amplitude" etc.

**Derivation:** Start with the equality of our ansatz and the asymptotic wavefunction:

$$e^{ikz} + f(\theta)\,\frac{e^{ikr}}{r} = \sum_{\ell=0}^{\infty} i^{\ell}\sqrt{4\pi(2\ell+1)}\frac{1}{2i}\left(\frac{e^{i(kr-\ell\pi/2+2\delta_\ell)} - e^{-i(kr-\ell\pi/2)}}{2i}\right)P_\ell(\cos\theta). \tag{37}$$

Using Eq. (30) for the incident plane wave, we can move the plane-wave to the right-hand side and use the asymptotic form of the spherical Bessel function ($j_\ell(kr) \sim \frac{1}{2i}\left(e^{i(kr-\ell\pi/2)} - e^{-i(kr-\ell\pi/2)}\right)$) to get

$$f(\theta)\,\frac{e^{ikr}}{r} = \sum_{\ell=0}^{\infty} i^{\ell}\sqrt{4\pi(2\ell+1)}\frac{1}{2i}\left(\frac{e^{i(kr-\ell\pi/2)}\left(e^{2i\delta_\ell}-1\right)}{2i}\right)P_\ell(\cos\theta). \tag{38}$$

Following Bart, we use the identity $e^{2i\delta_\ell} - 1 = 2ie^{i\delta_\ell}\sin\delta_\ell$ and $e^{-\ell\pi/2} = i^{-\ell}$ to simplify the right-hand side, and multiply by $e^{-ikr}/r$ to get

$$f(\theta) = \sum_{\ell=0}^{\infty}\sqrt{4\pi(2\ell+1)}\frac{1}{2i}\left(\frac{e^{i\delta_\ell}\sin\delta_\ell}{kr}\right)P_\ell(\cos\theta). \tag{39}$$

Plugging this into the expression for the total cross section, we find

$$\sigma = \int d\Omega\,|f(\theta)|^2 = \frac{4\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)\sin^2\delta_\ell, \tag{40}$$

where we used $\int_{-1}^{1} P_\ell(x)P_{\ell'}(x)\,dx = \frac{2}{2\ell+1}\,\delta_{\ell\ell'}$ to perform the angular integration. This is the famous **partial wave expansion** of the total cross section. Note how the orthogonality of the Legendre polynomials has simplified the angular integration, leaving us with a sum over $\ell$ of the contributions from each partial wave. That is: different $\ell$ amplitudes do not interfere with each other in the total cross section, since they are orthogonal functions of $\theta$.

There are special names for the first few partial waves: $\ell = 0$ is called the **s-wave**, $\ell = 1$ is the **p-wave**, $\ell = 2$ is the **d-wave**, and so on. In many cases, especially at low energies, the scattering is dominated by the s-wave, and the higher partial waves can be neglected. This is because the effective potential for each partial wave includes a centrifugal barrier term $\ell(\ell+1)/r^2$, which suppresses the contribution of higher $\ell$ values at low energies.

Another way to see this is to study the velocity dependence of the phase shifts $\delta_\ell$ at low energies. Since we are working in a non-relativistic theory, the velocity $v$ of the particle is related to the wavenumber $k$ by $v = k/m$. At low energies, the phase shifts typically scale as $\delta_\ell \sim k^{2\ell+1} \sim v^{2\ell+1}$, which means that the s-wave ($\ell = 0$) dominates as $v \to 0$. It is easy to understand this physically. At low velocities, the particle has a long de Broglie wavelength $\lambda = 2\pi/k$, which is much larger than the range $a$ of the potential. The particle effectively sees the potential as a point-like scatterer, and the scattering is dominated by the s-wave, which has no angular momentum barrier and can easily interact with the potential. At higher velocities, the particle can resolve the structure of the potential and higher partial waves can contribute significantly to the scattering process. If the potential is sufficiently strong, it can even support bound intermediate states, which manifest as resonances in the scattering amplitude at specific energies. In this case, the phase shift $\delta_\ell$ will rapidly change by $\pi/2$ as the energy passes through the resonance, leading to a large enhancement of the scattering amplitude and cross section at that energy.

### 1.5 Unitarity

Looking at Eq. (40), we see that the contribution of each partial wave to the total cross section is proportional to $\sin^2\delta_\ell$. Since $\sin^2\delta_\ell \leq 1$, this means that the contribution of each partial wave is bounded by

$$\sigma_\ell = \frac{4\pi}{k^2}(2\ell+1)\sin^2\delta_\ell \leq \frac{4\pi}{k^2}(2\ell+1). \tag{41}$$

This is a consequence of probability conservation, which requires that the scattering amplitude cannot grow without bound. In particular, the s-wave contribution is bounded by $\sigma_0 \leq 4\pi/k^2$, which is known as the **unitarity bound** for s-wave scattering. This bound has important implications for the behavior of scattering amplitudes at low energies, and it is a fundamental constraint that any physical scattering amplitude must satisfy.

For greater values of $\ell$, the unitarity bound becomes less stringent, since the contribution of higher partial waves is suppressed by the centrifugal barrier, but it may be useful if there are selection rules at play that forbid lower partial waves. For example, if the scattering particles are identical fermions, the s-wave contribution is forbidden by the Pauli exclusion principle,[3] and the p-wave contribution is bounded by $\sigma_1 \leq 12\pi/k^2$.

Note that the scale of the cross section may be set by the range of the potential, $\sigma \sim 4\pi a^2$, rather than by the momentum of the particle as in the unitarity bound. Indeed, if $a \ll 1/k$, the cross section is much smaller than the unitarity bound, so the unitarity bound, while useful as a bound, may not always be a good *estimate* for the cross section.

### 1.6 Optical Theorem

Let us study the forward direction now, $\theta = 0$. From the partial wave expansion, we have

$$f(0) = \frac{1}{k}\sum_{\ell=0}^{\infty}(2\ell+1)e^{i\delta_\ell}\sin\delta_\ell. \tag{42}$$

A simple trick shows that the imaginary part of $f(0)$ is related to the total cross section:

$$\operatorname{Im} f(0) = \frac{1}{k}\sum_{\ell=0}^{\infty}(2\ell+1)\sin\delta_\ell\sin\delta_\ell = \frac{k}{4\pi}\sigma. \tag{43}$$

This is the **optical theorem**, usually written as

$$\boxed{\sigma = \frac{4\pi}{k}\operatorname{Im} f(0).} \tag{44}$$

It relates the total cross section, which is an integral over all scattering angles, to the imaginary part of the forward scattering amplitude. It is a direct consequence of probability conservation and the unitarity of the scattering matrix, and it provides a powerful consistency check for any scattering amplitude calculation. In particular, it implies that if the scattering amplitude has a nonzero imaginary part in the forward direction, there must be a nonzero total cross section (some scattering happened).

#### 1.6.1 Optical Theorem Again

Let us return to Eq. (11) and derive the optical theorem directly from probability conservation and the interference current. The total probability current for $\psi = \psi_i + \psi_s$ splits into three pieces:

$$\mathbf{j} = \mathbf{j}_i + \mathbf{j}_s + \mathbf{j}_{\text{int}}, \tag{45}$$

[3] You can understand this by considering the symmetry of the wavefunction under particle exchange. If the wavefunction is antisymmetric, the s-wave component must vanish since a spherically-symmetric wavefunction is symmetric under exchange.

where the interference term is

$$\mathbf{j}_{\text{int}} = \frac{1}{2mi}\left(\psi_i^* \nabla \psi_s + \psi_s^* \nabla \psi_i - \psi_i \nabla \psi_s^* - \psi_s \nabla \psi_i^*\right). \tag{46}$$

Probability conservation requires that the total flux through a large sphere of radius $r$ vanishes:

$$\Phi_i + \Phi_s + \Phi_{\text{int}} = 0, \qquad \text{where} \quad \Phi_X = \int (\mathbf{j}_X \cdot \hat{r})\, r^2\, d\Omega. \tag{47}$$

The incident plane wave has zero net flux through any closed surface, $\Phi_i = 0$, and we already found $\Phi_s = (k/m)\sigma$. We therefore need $\Phi_{\text{int}} = -(k/m)\sigma$.

To compute $\Phi_{\text{int}}$, we evaluate the radial component of $\mathbf{j}_{\text{int}}$ at large $r$. Using $\partial_r \psi_i = ik \cos\theta\, e^{ikr\cos\theta}$ and $\partial_r \psi_s \simeq ikf\, e^{ikr}/r$ hwere we drop subleading $1/r^2$ terms, we get

$$j_{\text{int},r} = \frac{k}{mr}(1+\cos\theta)\,\text{Re}\big[f(\theta)\, e^{ikr(1-\cos\theta)}\big]. \tag{48}$$

Integrating over the sphere

$$\Phi_{\text{int}} = \frac{\pi k r}{m} \int_0^2 (2-u)\left[f(u)\, e^{ikru} + f^*(u)\, e^{-ikru}\right] du, \tag{49}$$

where $u = 1 - \cos\theta$. For $kr \gg 1$ the exponentials oscillate rapidly, so only the most forward points around $u = 0$ contribute. Integration by parts gives, to leading order,

$$\int_0^2 (2-u)\, f(u)\, e^{ikru}\, du = \frac{-2f(0)}{ikr} + \mathcal{O}\left(\frac{1}{(kr)^2}\right), \tag{50}$$

where the boundary at $u = 2$ vanishes because of the $(2-u)$ prefactor. The conjugate integral gives $+2f^*(0)/(ikr)$, so that

$$\Phi_{\text{int}} = \frac{\pi k r}{m} \cdot \frac{2}{ikr}\left[-f(0) + f^*(0)\right] = -\frac{4\pi}{m}\,\text{Im}\, f(0). \tag{51}$$

Setting $\Phi_s + \Phi_{\text{int}} = 0$, we recover the optical theorem:

$$\sigma = \frac{4\pi}{k}\,\text{Im}\, f(0). \tag{52}$$

The interpretation is that the scattered wave removes particles from the forward beam by destructively(!) interfering with the incident plane wave. The total cross section measures this "shadow" cast by the scatterer, and the optical theorem tells us this shadow is just $\text{Im}\, f(0)$. Note that for $\theta \neq 0$ the interference between the incident and scattered waves averages to zero as $kr \to \infty$.

**The index of refraction.** There is a nice connection between the optical theorem and the classical index of refraction in optics. Consider a plane wave $e^{ikz}$ passing through a thin slab of thickness $\delta z$ containing $N$ identical scatterers per unit volume. Each scatterer produces an outgoing spherical wave with amplitude $f(\theta)$. In the forward direction, the scattered waves from all scatterers in the slab add coherently (they are in phase), while at other angles the contributions from randomly distributed scatterers cancel by destructive interference. The wave after the slab is then the incident wave plus the sum of all forward-scattered waves. Summing over all scattering points in a transverse slice (imagine an infinitely wide plane) and using the same stationary-phase argument as above (only the forward direction $\theta = 0$ survives the rapidly oscillating integrals), one can show that

$$\psi(z + \delta z) \approx e^{ikz} \left(1 + \frac{2\pi i N}{k} f(0)\, \delta z\right) \approx e^{ikz}\, e^{i(2\pi N f(0)/k)\, \delta z}. \tag{53}$$

After propagating through a macroscopic distance $L$ of the medium (by stacking many thin slabs), the wave accumulates a phase:

$$\psi(L) = e^{in\,k\,L}, \quad \text{with} \quad n = 1 + \frac{2\pi N}{k^2} f(0), \tag{54}$$

where $n$ is the refractive index of the medium, expressed entirely in terms of the forward scattering amplitude $f(0)$ and the number density $N$ of scatterers. This is why light slows down in a medium. The forward-scattered waves interfere with the incident wave, slowing down its phase velocity.

Since $f(0)$ is in general complex, the refractive index is also complex: $\mathrm{Re}[f(0)]$ controls the phase velocity (refraction), while $\mathrm{Im}[f(0)]$ controls the attenuation (absorption). Indeed, the imaginary part of $n$ produces an exponentially decaying amplitude $e^{-\kappa L}$ and the intensity falls as

$$e^{-2\kappa L}, \quad \kappa = 2\pi N \mathrm{Im}[f(0)]/k^2 = N\sigma/2. \tag{55}$$

Comparing this with the probability of a particle to pass through a medium without scattering, $e^{-N\sigma L}$, we see that the cross section in this attenuation picture is just

$$\sigma = \frac{2\kappa}{N} = \frac{4\pi}{k}\, \mathrm{Im}\, f(0), \tag{56}$$

recovering yet again the **optical**(!) theorem.